

Unburying Martian History: Using ballistic physics and debris-flow models to constrain the thickness of giant impact ejecta on Mars

Abstract

The giant Noachian impacts that formed the Hellas, Argyre, Isidis, and Utopia basins blanketed the Martian surface with millions of cubic kilometers of ejecta. It is often assumed that ejecta from giant impacts buried pre-Noachian craters and topography, which would explain why the oldest terrain on Mars crater-dates to the early Noachian. However, the thickness distribution of ejecta layers from giant Martian impacts and their interaction with the pre-existing topography is not well understood, and it is not certain that basin ejecta would be thick enough to globally erase pre-Noachian craters. We propose to utilize ballistic physics to estimate global ejecta thickness on Mars and use debris flow modeling to simulate emplacement of massive ejecta blankets on modeled pre-Noachian terrain. This modeling program will constrain the extent to which ejecta from giant Noachian impacts could have buried pre-Noachian topography, enabling us to determine whether additional endogenic processes are required to erase the pre-Noachian record. As new data from rovers and orbiters offer us increasingly detailed views of Mars's surface, it is imperative that we understand exactly how the giant impact layer shaped the landscape and whether any remnants of pre-Noachian topography still exist so that we can correctly interpret Mars' geologic record. Investigating how giant impacts affect preexisting topography has important implications for other bodies that have been shaped by giant impacts and their ejecta, such as the Moon, Mercury, and Callisto.

- **Scientific Justification**

Investigating the distribution of giant impact ejecta on Mars.

In the violent collisional environment of the early solar system, giant impacts played an integral role in shaping planetary surfaces. Four giant impacts are known to have occurred early in Mars' history during the Noachian epoch: the Hellas impact, which marks the start of the Noachian; along with Argyre, Isidis, and Utopia. These massive impacts undoubtedly generated large amounts of ejecta that now blankets the Martian surface. Ejecta from Hellas and the other Noachian impacts is thought to have almost completely obscured the pre-Noachian geologic record (Carr & Head), as the oldest terrain on Mars dates to the early Noachian (Tanaka, 1986). A simple estimate of the combined thickness of the four large impacts using a lunar basin ejecta thickness model (Petro and Pieters, 2008) indicates that even the thinnest regions of the global ejecta layer are 300-400 meters thick. Observations and model results both suggest that ejecta from Hellas, Isidis, Argyre, and Utopia is abundant across the surface of Mars and has played a significant role in shaping the Martian geologic record.

However, the thickness distribution of giant impact ejecta on Mars is poorly constrained, and the effect of underlying topography on ejecta blanket morphology is not well understood. We cannot directly observe the properties of ancient basins' ejecta blankets because they have been heavily modified by later impacts, water-driven erosion, and volcanism (Robbins et al., 2013). It is unclear whether ejecta from Noachian impacts is thick enough globally to have fully erased the pre-Noachian geologic record (Petro and Pieters, 2008); it is possible that subtle remnants of the pre-Noachian topography are still expressed on the surface (Frey, 2006), or that other processes at work during the pre-Noachian played a role in degrading the landscape. As rovers and spacecraft return increasingly comprehensive and high-quality data from the Martian surface, it is vital that we better understand the nature of Noachian impact ejecta so that we can properly contextualize observations.

We propose to combine ballistic ejecta modeling with debris flow models to simulate giant ejecta blankets on Mars, and to constrain the thickness of giant impact ejecta and investigate its interactions with pre-existing topography. This work aligns with priority science questions outlined by the Planetary Science Decadal Survey Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023 – 2032 (National Academies of Sciences, 2023), which guide the priorities of the NASA FINESST Planetary Science Research Program. We will use ballistic modeling to create the **first global map of ejecta thickness on Mars**, which will be made publicly available so that other investigators can use it to interpret the Martian geologic record. Our debris flow modeling will produce **a new framework for understanding the interactions between basin ejecta and underlying topography**, which will be essential for studying planetary bodies throughout the solar system that have experienced giant impacts.

The distribution of giant impact ejecta on Mars is poorly constrained.

The giant Noachian impacts Hellas, Argyre, Isidis, and Utopia generated millions of cubic kilometers of ejecta that blankets the Martian surface, mixing with local surface material and burying pre-Noachian terrain. However, the combined thickness of ejecta from major impacts on Mars and the extent to which basin ejecta erased pre-Noachian

topography have not yet been extensively studied. Large basins on Mars rarely maintain a visible ejecta blanket (Carr and Head, 2010; Robbins et al., 2013), so it is difficult to determine the extent of an ejecta blanket beyond the immediate vicinity of the basin rim. Though it is often assumed that basin ejecta destroyed the pre-Noachian geologic record, some studies have proposed that some long-wavelength pre-Noachian topographic features are still subtly visible but buried by Noachian basin ejecta (Frey, 2006; Frey et al., 2001). Investigating the nature of giant impact ejecta will reveal whether buried pre-Noachian topography may still be visible, and whether processes besides giant impacts (e.g. erosion, volcanism) played a role in degrading pre-Noachian topography to the observed extent.

There is not yet a widely accepted estimate for the thickness distribution of ejecta from the giant Noachian impacts or a detailed model for how basin ejecta of Martian composition alters re-existing topography. Current estimates for the thickness of basin ejecta as a function of distance from the impact are largely based on Lunar impacts (Petro and Pieters, 2008; Pike, 1974), and such estimates must be updated to account for Mars' size and surface composition. This proposed program will generate a new map of estimated ejecta thickness from the four giant Noachian impacts, and will generate examples of ejecta draped over pre-existing cratered terrain so that we can understand the extent to which ejecta modified pre-Noachian topography.

This proposal: ballistic and debris flow modeling of giant impact ejecta on Mars.

This study targets the Martian surface, specifically the role of basin ejecta in shaping the topography and burying the ancient geologic record. The NASA FINESST Planetary Science Research Program solicits proposals for research that aligns with the priority questions outlined in the Planetary Science Decadal Survey Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023 – 2032 (OWL). Our proposed work aligns with OWL Priority Science Questions relating to impacts and planetary surfaces, and therefore our research is well-suited for FINESST support.

We propose to use ballistic projectile modeling to predict ejecta thickness as a function of distance from the center of a giant impact on Mars, and then to use debris-flow modeling to predict how ejecta modified the pre-Noachian landscape. The ballistic modeling will be calibrated to account for Mars' size and surface properties, rather than relying on Lunar estimates. We will use this new ejecta thickness vs. distance relationship to estimate the combined thickness of ejecta from the four giant Noachian impacts, improving upon the preliminary map shown in Figure 1A. Once we have estimated ejecta thickness at different regions of Mars, we will use a granular debris flow model to simulate the appearance of debris layers of various thicknesses blanketing heavily-cratered pre-Noachian topography. Pre-Noachian topography will be simulated using a Martian cratering model (Fulford et al., 2024). Preliminary results from using a simple airfall model to emplace an ejecta layer on modeled pre-Noachian topography (Fig. 1B) show that pre-Noachian craters are altered but still visible beneath the ejecta blanket, contradicting the widely-held assumption that giant basin ejecta fully erased the pre-Noachian geologic record. Utilizing a more complex debris-flow model will generate a more accurate representation of ejecta blanketing a surface, allowing us to verify preliminary results.

The major product generated by this research will be the **first** global map of estimated combined ejecta thickness from Hellas, Isidis, Argyre, and Utopia across the

Martian surface. This map, and accompanying models of ejecta draped over Martian terrain, will vastly improve our understanding of the role of giant impact ejecta in Mars' evolution and will provide essential context for future geologic studies.

Broader implications for Martian geology and planetary surface evolution.

Giant impacts have occurred throughout the solar system and hugely alter the target body, and therefore understanding the behavior of basin ejecta is essential for studying the surface of a heavily cratered terrestrial body like Mars. Our work will advance the subfield of Martian surface geology by determining the contribution of giant impact ejecta to Martian topography and constraining the extent to which pre-Noachian topography is still expressed on the surface.

Understanding basin ejecta distribution is increasingly vital for interpreting groundbreaking data returned by rovers and orbiters, such the Perseverance Observations of ancient rock thought to be ejecta from Isidis (Quantin-Nataf et al., 2026). Our study will be vital for answering several key questions posed by the OWL related to impacts on terrestrial bodies (National Academies of Sciences, 2023), including Q4.3 (“How did collisions affect the geological, geophysical, and geochemical evolution of planetary bodies?”) and Q5.5b (“How do the physics and mechanics of impacts produce disruption of and cratering on planetary bodies?”). Our study will be the first to create maps of basin ejecta thickness on Mars and will introduce a new framework for studying interactions between ejecta and the underlying terrain, both of which will be made available to the wider community to be utilized in innovative future investigations of Mars and other bodies affected by giant impacts.

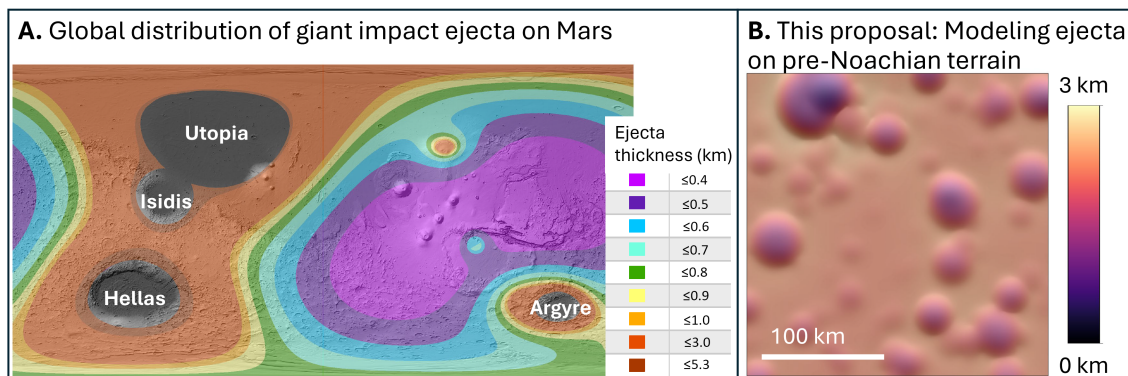


Figure 1: The ejecta generated by Mars’s four giant Noachian impacts blankets the southern highlands and obscures the pre-Noachian geologic record. Panel A displays the estimated thickness of combined ejecta from these four impacts across the Martian surface. The thinnest region is 300-400 m. Panel B shows results from a simple airfall model used to emplace ~350 m of debris on a pre-Noachian cratered landscape generated using a cratering model (Fulford et al., 2024). The craters are altered but still visible. Giant basin ejecta is widespread on Mars but currently poorly explained, so we must form a detailed understanding of how ejecta interacts with pre-existing topography. We propose to combine ballistic physics with a debris flow model to constrain the thickness of ejecta across the Martian surface and to predict the appearance of an ejecta layer blanketing heavily-cratered terrain, enabling us to determine whether any pre-Noachian topography still exists on Mars and offering new insight into Mars’s evolution from the pre-Noachian to the early Noachian.

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